

Scope and features of cloud simulations

- Data centres
- Load balancing
- Creation and execution of cloudlets
- Resource provisioning
- Scheduling of tasks
- Storage and cost factors
- Energy optimisation ... and many others

Cloud simulation tools and plugins

- CloudSim
- CloudAnalyst
- GreenCloud
- iCanCloud
- MDCSim
- NetworkCloudSim
- VirtualCloud
- CloudMIG Xpress
- CloudAuction
- CloudReports
- RealCloudSim
- DynamicCloudSim
- WorkFlowSim

in the domain of cloud computing.

It's important to specify here that we should try to get all the research papers from journals related to the topic being researched rather than from journals of all streams. If the researcher is working on cloud computing, then the papers should be taken from reputed journals in the cloud computing domain rather than the classical journals of computer science.

The specific international journals of cloud computing include:

- Journal of Cloud Computing – SpringerOpen Journal
- IEEE Transactions on Cloud Computing

In case other international journals covering cloud computing are consulted, integrity in terms of indexing should be verified very carefully. Research papers carried in Science Citation Indexed (SCI) Journals should be verified from the URL <http://ip-science.thomsonreuters.com/cgi-bin/jrnlist/jfoptions.cgi?PC=K> of Thomson Reuters to avoid any problems. Currently, many universities strictly direct their research scholars to publish their papers only in SCI Journals.

International journals that have indexing with SCI and Scopus should be analysed. Nowadays, many international journals claim to have SCI, SCI-Expanded or Scopus indexing, but the researcher should check their authenticity. It is also suggested that researchers check the validity of the Impact Factor from the website of Thomson Reuters.

After detailed study of the research papers from the latest journals, the problem or gap in the classical algorithms is pointed out. In every research paper, the algorithm, results and conclusion, as well as the future work gives an idea of how the proposed work can be improved. The algorithm mentioned in the research paper can be integrated with any meta heuristic technique, including Simulated Annealing, Ant Colony Optimisation, the HoneyBee Algorithm or any other. The meta heuristic techniques can give effective results from any classical algorithm mentioned in the base paper.

Once the problem is found, the synopsis or research proposal is prepared, which is one of the most important tasks in the whole research path, and hence should be done very carefully. Generally, the synopsis is written by the research scholars or supervisors in a very haphazard and careless manner. But once the synopsis

is approved by the university, it becomes very dangerous if the students or guides themselves are not too clear about all the points mentioned in the research proposal. The research committee of the university focuses on the objectives and hypothesis before approving the synopsis. Students or guides should draft hypotheses and specify the objectives of the research with utmost care. Generally, students/guides give very hypothetical objectives to impress the research committee and after approval of the synopsis, it might become very difficult to achieve all the grand objectives mentioned in the research proposal, in order to complete the degree.

Simulation of a research solution in cloud simulators

Cloud service providers charge their clients depending upon the space or service provided to them. In R&D, it is not always possible to have the actual cloud infrastructure for performing the experiments. For any research scholar, academician or scientist, it is not feasible to hire cloud services and then execute their algorithms or implementations, every time. For the purpose of R&D and testing, open source libraries are available, using which, the feel of cloud services and executions can be experienced. Nowadays, in the research market, cloud simulators are widely used by research scholars and practitioners without paying fees to any cloud service provider.

Using cloud simulators, researchers can execute their algorithmic approaches on a software-based library and can get the results for different parameters including energy optimisation, security, integrity, confidentiality, bandwidth, power and many others.

Tasks performed by using cloud simulators

Cloud simulators help to perform the following tasks:

- Modelling and simulation of large scale cloud computing data centres
- Modelling and simulation of virtualised server hosts, with customisable policies for provisioning host resources to virtual machines
- Modelling and simulation of energy-aware computational resources
- Modelling and simulation of data centre network topologies and message-passing applications
- Modelling and simulation of federated clouds
- Dynamic insertion of simulation elements, stopping and resuming of simulation
- User-defined policies for allocating hosts to virtual machines, and policies for allocation of host resources to virtual machines 

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